

## LUHANKA JUDINSALO, Finland

WMO: 027650

Lat: 61.704N

Lon: 25.505E

Elev: 273

StdP: 14.55

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -11.4      | -4.4 | -15.3                         | 2.4 | -11.2 | -7.8 | 3.6 | -3.1 | 32.7                  | 26.2 | 29.1 | 25.0 | 4.9                   | 10   | 0.667 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.9                   | 79.0            | 66.7 | 75.3 | 64.9 | 72.1 | 63.2 | 68.8                | 75.7 | 67.0 | 73.4 | 64.7 | 70.7 | 6.7                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.3                            | 98.0 | 72.8 | 64.4 | 91.3 | 71.3 | 62.0 | 83.9 | 68.7 | 33.2          | 75.4 | 31.7 | 73.8 | 29.9 | 70.6 | 74.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |      | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|------|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |      | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max  | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h)  | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| 27.0              | 23.8 | 21.0 | DB                         | -12.4 | 80.9               | 11.7 | 5.0                                                | -20.9 | 84.5       | -27.7 | 87.4       | -34.3 | 90.2       | -42.8 | 93.8 | (4) |
|                   |      |      | WB                         | -12.8 | 69.8               | 11.6 | 2.6                                                | -21.1 | 71.7       | -27.9 | 73.2       | -34.5 | 74.7       | -42.9 | 76.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                               |                                                                               | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                               |                                                                               | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg                                                                         | 41.7   | 20.3  | 22.6  | 27.5  | 36.7  | 49.6  | 58.0  | 64.4  | 61.5  | 53.5  | 42.2  | 34.9  | 28.3 |      |
|                                                                               | DBStd                                                                         | 16.80  | 11.43 | 11.68 | 8.18  | 6.33  | 7.23  | 6.08  | 6.27  | 4.96  | 5.23  | 5.92  | 7.35  | 9.89 |      |
|                                                                               | HDD50                                                                         | 4288   | 921   | 767   | 697   | 400   | 97    | 6     | 0     | 0     | 28    | 246   | 453   | 673  |      |
|                                                                               | HDD65                                                                         | 8599   | 1386  | 1187  | 1162  | 849   | 477   | 223   | 91    | 132   | 345   | 706   | 903   | 1138 |      |
|                                                                               | CDD50                                                                         | 1272   | 0     | 0     | 0     | 1     | 85    | 246   | 447   | 355   | 133   | 5     | 0     | 0    |      |
|                                                                               | CDD65                                                                         | 109    | 0     | 0     | 0     | 0     | 1     | 13    | 73    | 22    | 0     | 0     | 0     | 0    |      |
|                                                                               | CDH74                                                                         | 444    | 0     | 0     | 0     | 0     | 4     | 59    | 326   | 55    | 0     | 0     | 0     | 0    |      |
|                                                                               | CDH80                                                                         | 62     | 0     | 0     | 0     | 0     | 0     | 8     | 48    | 6     | 0     | 0     | 0     | 0    |      |
| Wind                                                                          | WSAvg                                                                         | 9.5    | 9.6   | 9.6   | 9.8   | 8.2   | 8.4   | 9.1   | 8.2   | 8.9   | 9.3   | 11.0  | 10.7  | 11.2 |      |
| Precipitation                                                                 | PrecAvg                                                                       | 24.3   | 1.9   | 1.4   | 1.2   | 1.3   | 1.6   | 2.5   | 3.2   | 2.8   | 2.0   | 2.4   | 2.1   | 1.9  |      |
|                                                                               | PrecMax                                                                       | 31.8   | 3.5   | 3.1   | 2.4   | 2.3   | 4.2   | 4.1   | 6.5   | 5.4   | 5.5   | 5.2   | 4.4   | 3.7  |      |
|                                                                               | PrecMin                                                                       | 20.1   | 0.4   | 0.1   | 0.1   | 0.2   | 0.3   | 1.1   | 0.6   | 0.6   | 0.4   | 0.6   | 0.2   | 0.5  |      |
|                                                                               | PrecStd                                                                       | 2.9    | 0.8   | 0.7   | 0.5   | 0.6   | 0.9   | 0.8   | 1.4   | 1.2   | 1.1   | 1.1   | 0.9   | 0.8  |      |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                          | DB     | 38.6  | 41.7  | 46.1  | 60.5  | 73.3  | 80.6  | 84.5  | 80.4  | 67.5  | 55.5  | 51.4  | 45.9 |      |
|                                                                               |                                                                               | MCWB   | 37.0  | 37.2  | 38.9  | 48.4  | 56.6  | 61.3  | 68.9  | 69.0  | 60.7  | 52.4  | 48.5  | 44.5 |      |
|                                                                               | 2%                                                                            | DB     | 36.4  | 38.1  | 41.8  | 52.9  | 69.1  | 74.6  | 80.7  | 74.7  | 64.4  | 53.5  | 47.9  | 41.7 |      |
|                                                                               |                                                                               | MCWB   | 35.5  | 35.9  | 36.9  | 43.9  | 56.2  | 62.0  | 67.2  | 65.6  | 58.8  | 51.2  | 46.5  | 40.1 |      |
|                                                                               | 5%                                                                            | DB     | 34.7  | 35.7  | 39.1  | 49.1  | 65.2  | 70.6  | 77.7  | 71.5  | 62.2  | 52.2  | 45.7  | 39.5 |      |
|                                                                               |                                                                               | MCWB   | 34.0  | 34.4  | 34.8  | 41.9  | 54.4  | 59.6  | 66.7  | 63.1  | 57.7  | 50.0  | 44.5  | 38.1 |      |
|                                                                               | 10%                                                                           | DB     | 33.3  | 34.1  | 36.8  | 45.2  | 61.3  | 67.4  | 74.4  | 69.0  | 60.3  | 50.3  | 43.7  | 37.9 |      |
|                                                                               |                                                                               | MCWB   | 32.7  | 33.4  | 33.9  | 39.6  | 52.2  | 58.8  | 65.3  | 61.8  | 56.5  | 48.2  | 42.6  | 36.8 |      |
|                                                                               | Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 37.6  | 37.8  | 40.4  | 50.3  | 60.9  | 67.3  | 71.2  | 71.1  | 61.9  | 53.3  | 48.5 | 44.9 |
|                                                                               |                                                                               |        | MCDB  | 38.4  | 40.3  | 43.9  | 54.8  | 67.6  | 75.0  | 79.4  | 78.4  | 65.8  | 54.4  | 50.4 | 46.1 |
| 2%                                                                            |                                                                               | WB     | 35.7  | 36.2  | 37.6  | 45.3  | 58.3  | 63.7  | 69.6  | 67.4  | 60.2  | 52.0  | 46.7  | 40.5 |      |
|                                                                               |                                                                               | MCDB   | 36.3  | 37.9  | 40.7  | 52.3  | 65.5  | 70.4  | 76.7  | 73.0  | 63.1  | 53.3  | 48.0  | 41.4 |      |
| 5%                                                                            |                                                                               | WB     | 34.1  | 34.8  | 35.9  | 42.7  | 56.1  | 61.8  | 68.3  | 65.0  | 58.8  | 50.6  | 44.7  | 38.4 |      |
|                                                                               |                                                                               | MCDB   | 34.5  | 35.8  | 38.2  | 48.0  | 63.4  | 68.4  | 75.2  | 70.0  | 61.5  | 51.8  | 45.7  | 39.3 |      |
| 10%                                                                           |                                                                               | WB     | 32.9  | 33.5  | 34.6  | 40.3  | 53.1  | 59.9  | 66.7  | 63.0  | 57.5  | 48.3  | 42.9  | 36.9 |      |
|                                                                               |                                                                               | MCDB   | 33.4  | 34.1  | 36.2  | 44.2  | 60.7  | 66.5  | 73.1  | 67.6  | 59.9  | 50.0  | 43.6  | 37.8 |      |
| Mean Daily<br>Temperature<br>Range                                            |                                                                               | 5% DB  | MDBR  | 8.4   | 9.0   | 11.6  | 12.9  | 14.1  | 11.6  | 10.9  | 9.5   | 7.8   | 6.3   | 5.4  | 6.0  |
|                                                                               |                                                                               |        | MCDBR | 5.9   | 7.7   | 12.7  | 19.0  | 20.1  | 16.6  | 14.6  | 13.0  | 10.5  | 7.4   | 6.9  | 6.7  |
|                                                                               | MCWBR                                                                         |        | 5.6   | 6.5   | 9.1   | 11.0  | 9.5   | 7.2   | 6.4   | 6.6   | 6.2   | 5.9   | 6.4   | 6.5  |      |
|                                                                               | 5% WB                                                                         | MCDBR  | 5.8   | 7.6   | 10.5  | 17.0  | 18.7  | 14.2  | 13.0  | 11.1  | 9.4   | 7.0   | 6.9   | 6.6  |      |
|                                                                               |                                                                               | MCWBR  | 5.6   | 6.7   | 8.1   | 10.8  | 9.9   | 6.9   | 6.7   | 6.8   | 6.1   | 6.7   | 6.5   | 6.5  |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub                                                                          | 0.247  | 0.292 | 0.320 | 0.359 | 0.356 | 0.352 | 0.376 | 0.361 | 0.338 | 0.319 | 0.269 | 0.357 |      |      |
|                                                                               | taud                                                                          | 2.064  | 2.147 | 2.228 | 2.353 | 2.435 | 2.475 | 2.425 | 2.456 | 2.512 | 2.412 | 2.162 | 2.600 |      |      |
|                                                                               | Ebn at Noon                                                                   | 140    | 210   | 249   | 264   | 274   | 278   | 266   | 261   | 247   | 205   | 131   | 92    |      |      |
|                                                                               | Edh at Noon                                                                   | 12     | 22    | 29    | 32    | 32    | 31    | 32    | 29    | 23    | 17    | 11    | 7     |      |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg                                                                        | 69     | 247   | 713   | 1164  | 1570  | 1656  | 1596  | 1238  | 719   | 301   | 80    | 36    |      |      |
|                                                                               | RadStd                                                                        | 12     | 57    | 88    | 126   | 176   | 129   | 162   | 143   | 94    | 55    | 15    | 5     |      |      |

## Historical Trends

|                         |       | DBAvg  |        | Heating |       | Cooling |       | Degree-Days |       |       |      |
|-------------------------|-------|--------|--------|---------|-------|---------|-------|-------------|-------|-------|------|
|                         |       | 99% DB | 99% DP | 1% DB   | 1% WB | 1% DP   | HDD50 | HDD65       | CDD50 | CDD65 |      |
| (d)                     | (e)   | (f)    | (g)    | (h)     | (i)   | (j)     | (k)   | (l)         | (m)   |       |      |
| Station Only            | N/A   | N/A    | N/A    | N/A     | N/A   | N/A     | N/A   | N/A         | N/A   | N/A   | (46) |
| Regional (56 neighbors) | +0.88 | N/A    | N/A    | N/A     | N/A   | N/A     | -266  | N/A         | N/A   | N/A   | (47) |

Nomenclature: See separate page